

September 12, 2026

To the Editor-in-Chief
Bioinformatics
Oxford University Press

Submission of an Application Note

DEploid-BEST: an optimised workflow for deconvolving mixed P. falciparum infections

Dear Editor,

Please find enclosed our manuscript entitled *DEploid-BEST: an optimised workflow for deconvolving mixed P. falciparum infections*, which we submit for consideration as an **Application Note** in *Bioinformatics*.

This work has not been published previously, is not under consideration for publication elsewhere, and, if accepted, will not be published elsewhere in substantially the same form.

Summary. As a state-of-the-art method for malaria parasite genome deconvolution, the program **DEploid** takes read counts from a mixed sample and estimates the number of strains, their relative proportions and their sequences from a set of reference strains. This output provides a unique opportunity for researchers to study the epidemiology of infectious diseases. However, the complexity of **DEploid** is $\mathcal{O}(lm^2)$, where m is the number of reference strains and l is the number of sites, so it does not scale well to reference panels with many strains. To address this, we present **DEploid-LASSO**, which can take thousands of reference strains and shrink and optimise the panel, together with a revised pipeline that we call **DEploid-BEST**, the latest best practice for deconvolving unknown mixed genomes.

Why this work fits *Bioinformatics*. Mixed-genome deconvolution is central to malaria surveillance and transmission-mapping, yet routine use of **DEploid** has been limited by its quadratic scaling with reference-panel size and by the need to pre-specify the number of strains. **DEploid-BEST** removes both bottlenecks: a LASSO step selects an effective reference panel (so run time is roughly constant as the panel grows from 10 to 1000 strains), and the strain number is recovered automatically while matching the accuracy of a correctly specified classical deconvolution. We believe the tool, its open-source implementation, and the benchmarking methodology will be of direct practical value to the bioinformatics and malaria-epidemiology communities that *Bioinformatics* serves, and we submit it as an Application Note describing readily usable software.

Availability and implementation. The open source implementation is freely available at <https://github.com/DEploid-dev/DEploid> under the conditions of the GPLv3 license. **DEploid-BEST** is implemented in C++ with an R interface.

Suggested reviewers.

Conflict of interest and funding. Software development and data analysis in this work were completed by S.J.Z. during a postdoc at the Big Data Institute. S.J.Z. is a principal data scientist at Roche; there is no conflict of interest. This work was supported by the Wellcome Trust grant [100956/Z/13/Z] to G.M.

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Thank you for your consideration. We look forward to hearing from you.

Sincerely,

Sha Joe Zhu
Gil McVean